

Revised Generation Interconnection System Impact Study Report

For Queue Project Q23-2305

Roanoke Rapids 138 kV
April 21, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2305. The Interconnection Customer proposes a 228.9 MW solar photovoltaic facility interconnecting at the Roanoke Rapids 138 kV substation in the North Carolina area. The request was received on March 28, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 676 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,514,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Tyler – Kewaunee Substation 138 KV
Contingency: Mccartney – N Tower Tap 345 KV
Loading: 122.8% (Rating: 227 MVA)
DFAX: 8.6%
Pre-project loading: 114.1%
2. Facility: Oyster Riv – Cedar Falls Wi 138 KV
Contingency: N Tower Tap – Haggin Substation 345 KV L
Loading: 122.2% (Rating: 212 MVA)
DFAX: 5.3%
Pre-project loading: 116.5%
3. Facility: Harvell – Oyster Riv 138 KV
Contingency: Haggin – North Tower Tap 345 KV
Loading: 121.7% (Rating: 191 MVA)
DFAX: 5.3%
Pre-project loading: 115.4%
4. Facility: Goldfield – Deen Pt 138 KV
Contingency: Mccartney – North Tower Tap Substation 345 KV
Loading: 117.2% (Rating: 266 MVA)
DFAX: 8.6%
Pre-project loading: 109.8%
5. Facility: Canon City – Villa Del Rey 3010 138 KV
Contingency: East Longmeadow Substation – Hard Rock 138 KV Line
Loading: 116.9% (Rating: 209 MVA)
DFAX: 8.4%
Pre-project loading: 107.7%

6. Facility: Canon City – Hard Rock 138 KV
Contingency: Villa Del Rey 3010 – Canon City 138 KV
Loading: 101.4% (Rating: 223 MVA)
DFAX: 8.0%
Pre-project loading: 93.2%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Kewaunee – Tyler 138 KV Line	Rebuild line with higher-capacity conductor	\$9,064,000	3% (\$283,000)
Oyster Riv – Cedar Falls Wi 138 KV	Rebuild line with higher-capacity conductor	\$2,678,000	4% (\$93,000)
Harvell – Oyster Riv 138 KV	Rebuild line with higher-capacity conductor	\$2,116,000	4% (\$74,000)
Deen Pt – Goldfield 138 KV	Rebuild line with higher-capacity conductor	\$2,400,000	3% (\$75,000)
Canon City – Villa Del Rey 3010 138 KV	Rebuild line with higher-capacity conductor	\$22,450,000	8% (\$1,833,000)
Hard Rock – Canon City 138 KV	Reconductor line	\$1,824,000	8% (\$156,000)
	Total allocated to project		\$2,514,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.